Paleomagnetism of Archean rocks from northwestern Ontario: IV. Burchell Lake granite, Wawa–Shebandowan Subprovince

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The late Archean Burchell Lake granite of the Shebandowan greenstone belt in northwestern Ontario has a characteristic natural remanent magnetization (NRM) resembling the type 1 NRM of the nearby Shelley Lake granite of the Quetico gneiss belt. Of 36 stably magnetized samples, 21 had predominantly normal polarity (1N) NRM and 15 had reverse polarity (1R). The mean direction based on stable end-point and vector subtracted directions is $D = 2.3°$ $I = 48.9°$ ($k = 23.9$, $\alpha_{95} = 10.0°$, $N = 10$ sites). Intersecting remagnetization circles gave a similar direction. The corresponding paleopole BL1 lies at 83.2°E, 71.1°N, near track 6 of the Laurentian apparent polar wander path around 2600 Ma. Although the Burchell Lake pluton is not dated radiometrically, neighbouring granites give K/Ar biotite ages of 2550–2600 Ma, and it is reasonable to suppose that the NRM dates from the time of intrusion or shortly thereafter. A type 2 NRM like that of the Shelley Lake granite was isolated in 12 samples. Its mean direction is $D = 80.9°$ $I = -11.1°$, but the precision is unacceptably low ($k = 7.7$).

Le granite du lac Burchell d'âge archéen supérieur appartient à la ceinture de roches vertes de Shebandowan du nord-ouest de l'Ontario et il comporte une aimantation rémanente naturelle (ARN) caractéristique qui ressemble à l'ARN de type 1 tout près du granite du lac Shelley, lequel fait partie de la ceinture gneissique de Quético. Parmi les 36 échantillons avec aimantation permanente, 21 affichent une polarité normale (1N) prédominante de leur ARN et 15 ont une polarité inversée (1I). La direction principale dérivée du point final stable et du vecteur de soustraction des directions est $D = 2{,}3°$ $I = 48{,}9°$, ($k = 23{,}9$, $\alpha_{95} = 10{,}0°$, $N = 10$ emplacements). L'intersection des cercles de réaimantation fournit une direction analogue. Le paléopôle correspondant BL1 se situe à 83,2°E, 71,1°N, proche de la courbe 6 de la dérive apparente des pôles à environ 2600 Ma. Quoique le pluton du lac Burchell ne soit pas daté radiométriquement, les granites voisins fournissent des âges établis par K/Ar sur biotite de 2550–2600 Ma, et il est raisonnable de prétendre que l'ARN procure un âge qui correspond à l'épisode d'instruction ou peu après. Une ARN de type 2 analogue à celle du granite du lac Shelley a été reconnue dans 12 échantillons. Sa direction moyenne est $D = 80{,}9°$, $I = -11{,}1°$ mais la précision ($k = 7{,}7$) est si faible qu'elle est inacceptable.

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Introduction

Previous papers in this series (Dunlop 1983, 1984; Dunlop *et al.* 1984; referred to in the rest of this paper as I, II, and III) have dealt with paleomagnetic and rock magnetic properties of the late Archean Wabigoon gabbro of the Wabigoon volcanic belt and the 2580 ± 20 Ma Shelley Lake granite of the Quetico gneiss belt, both of the Archean Superior Province in northwestern Ontario. The present paper describes similar studies carried out on the Burchell Lake granite of the Shebandowan–Wawa volcanic subprovince, which lies to the east of the Quetico Subprovince. A location map is given in II.

The natural remanent magnetizations (NRM's) of the Burchell Lake and Shelley Lake granites have much in common. Type 1 NRM, which was argued to be the primary 2580 Ma magnetization of the Shelley Lake body, is the characteristic Burchell Lake NRM. Type 2 NRM, a late Proterozoic remagnetization component found in about a dozen Shelley Lake samples, is identified in a few Burchell Lake samples, but is rather poorly defined, perhaps because NRM intensities tend to be an order of magnitude lower (geometric mean 45×10^{-6} emu/cm^3 (0.045 A/m)) than in the Shelley Lake granite (mean 300×10^{-6} emu/cm^3 (0.3 A/m)).

The Burchell Lake pluton has not been dated radiometrically. Granitic bodies a few miles distant have K/Ar biotite ages in the range 2550–2610 Ma (e.g., Giblin 1964, p. 21). The Shelley Lake type 1 NRM was shown in II and III to be in large part a primary thermoremanent magnetization (TRM), but this is probably not the case for many Burchell Lake samples. Although the Burchell Lake type 1 NRM and its corresponding paleomagnetic pole probably date from soon after intrusion around 2550–2600 Ma, the case is not as strong as it was for the Shelley Lake granite.

Geological setting

The Burchell Lake body is the largest of a number of porphyritic biotite granite plutons cutting a major rhyolite band on the western margin of the Shebandowan–Wawa greenstone belt. The rhyolite is flanked by steeply dipping mafic volcanics and probably forms the core of a northeast-trending anticline (Giblin 1964, p. 19). Primary structures have otherwise been generally obscured by Kenoran regional metamorphism. Compared with the often highly sheared metavolcanics and metasediments, the granitic plutons are undeformed. It will be assumed throughout this paper that the Burchell Lake pluton is post-kinematic and that paleomagnetic directions require no structural correction.

The Burchell Lake body is roughly circular in plan and has a diameter of about 4 km. It has a discontinuous syenitic border zone, but all paleomagnetic sites were taken in biotite granite, which forms the bulk of the body.

Sampling and measurement procedures

Fifty-three oriented samples were collected from seven sites on the southern margin of the intrusion near the former mine town of Burchell Lake on Highway 802 about 20 km southwest of Shebandowan, Ontario, and from three sites on the northern margin of the intrusion accessible from Highway 11. Samples were also taken from four sites in rhyolites and amphibolite schists along Highway 802 and in the Coldstream mine gabbro but none of the reheated country rocks gave stable magnetic results consistent with those of the granite. Samples were oriented by magnetic compass, with check readings taken 1–2 m above the outcrop.

Two or three 2.5 cm cylindrical specimens of each sample were alternating field (AF) demagnetized in the laboratory in

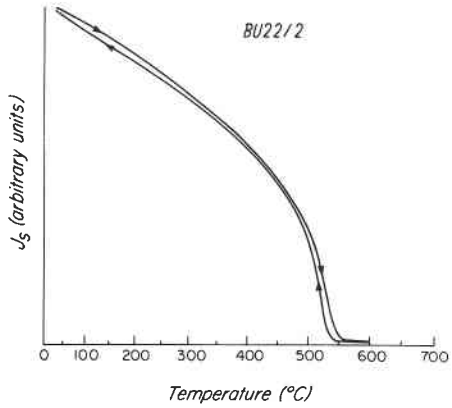


FIG. 1. A typical strong-field thermomagnetic curve for the Burchell Lake collection, showing reversible behaviour and a single Curie point caused by magnetite.

12 steps to 1000 Oe (0.1 T) using a Schonstedt single-axis demagnetizer. In addition, one specimen was thermally demagnetized, usually in 50°C steps to 500°C and then in smaller steps to 675°C using a noninductive furnace with Helmholtz coil field cancellation. Magnetizations were measured with Digico and Schonstedt spinner magnetometers.

Acceptance criteria were the same as in previous studies (I, II): (1) a stable grouping of magnetization vectors during AF or thermal cleaning with <20° between-specimen divergence; or (2) great-circle swings of the NRM vector in the course of demagnetization whether or not a stable end point was reached. Stable end points and subtracted vectors were defined by a minimum of four successive data points defining straight lines on an orthogonal vector plot. Great circles were usually fitted by eye from an equatorial stereographic projection, but in cases where the data were noisy and there was a margin of doubt, the fits were checked using the routine of Bailey and Halls (1984). Computed and estimated normals invariably agreed within 1−2°.

Thermomagnetic curves were measured in air in a field of 3000−5000 Oe (0.3−0.5 T) for 14 samples using a recording Curie balance. A full program of rock magnetic measurements (susceptibility, saturation hysteresis properties, AF demagnetization of synthetic remanences) was also carried out.

Summary of rock magnetic studies

The thermomagnetic curve shown in Fig. 1 is typical of the collection. Heating and cooling curves are identical, and almost pure magnetite with a Curie point near 550°C is the only important phase revealed. Thermal demagnetization shows that hematite carries a small but significant fraction of the NRM in some samples.

NRM intensities are <5% of saturation isothermal remanent magnetization (IRM_S) for all samples tested and <1% of IRM_S for half of these samples. These intensities are very low compared with laboratory TRM's produced by a field of 0.6 Oe (60 μT), which tend to be ≳5% of IRM_S.

The AF decay of NRM is soft and multidomain (MD) like, with median destructive fields of 100−150 Oe (10−15 mT). The Lowrie−Fuller test (Lowrie and Fuller 1971) gives MD-type results (NRM decaying more rapidly than IRM_S). The NRM thus seems to be carried largely by coarse MD magnetite that is greater than 15 μm in size (Bailey and Dunlop 1983). In many samples, the NRM was either produced by some less efficient process than TRM or has decayed significantly since it was first acquired.

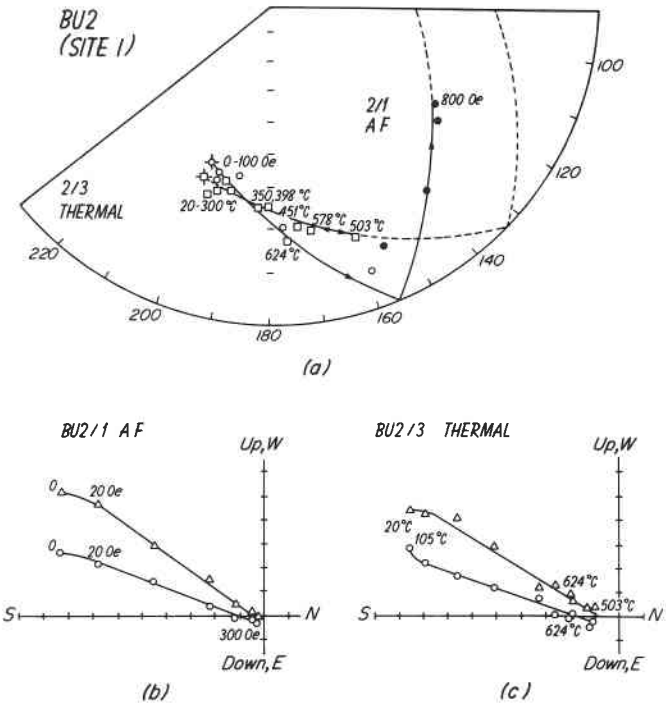


FIG. 2. Behaviour of the NRM vectors of two specimens of sample BU2 in the course of AF and thermal demagnetizations. (*a*) Stereographic projection of magnetization unit vectors. Open and solid symbols denote upward and downward vectors, respectively. Symbols marked with a vertical cross and labelled with a field or temperature range (e.g., 0−100 Oe, 20−300°C) indicate subtracted vectors over that range. (*b*, *c*) Orthogonal vector projections of AF and thermal demagnetization data, demonstrating the predominance of 1R magnetization in BU2. Circles represent horizontal-plane projections and triangles represent vertical-plane (Up vs. N or Up vs. E) projections.

Type 1 NRM results

Typical demagnetization characteristics

About one quarter of the Burchell Lake samples have a south or south-southwest declination, intermediate negative inclination NRM like that of BU2 in Fig. 2. This is the 1R remanence (type 1, reverse polarity). Most of the remanence in sample BU2 is carried by soft magnetite with coercivities (H_C's) mainly from 20 to 200 Oe (2−20 mT) (Fig. 2*b*) and blocking temperatures (T_B's) between 105 and 503°C (Fig. 2*c*). Between 503 and 624°C, the NRM vector partly retraces its great-circle swing towards the 1R grouping (Fig. 2*a*), as a result of the removal of a small 1N vector, 180° reversed to 1R (Fig. 2*c*). Above 624°C, the results become obscured by noise, but a certain fraction of 1R must be carried by hematite with T_B's in this range.

The important points to be drawn from Fig. 2 are that type 1 NRM is carried by both magnetite and hematite and that the two polarities of remanence appear to have mutually exclusive T_B ranges whose limits do not coincide with the Curie point of magnetite.

Sample BU5 (Fig. 3) provides an example of a predominantly 1N NRM. The 1N direction is about 20° shallower than the present Earth's field (PEF) direction. Only above 597°C does a small antiparallel 1R vector carried by hematite begin to

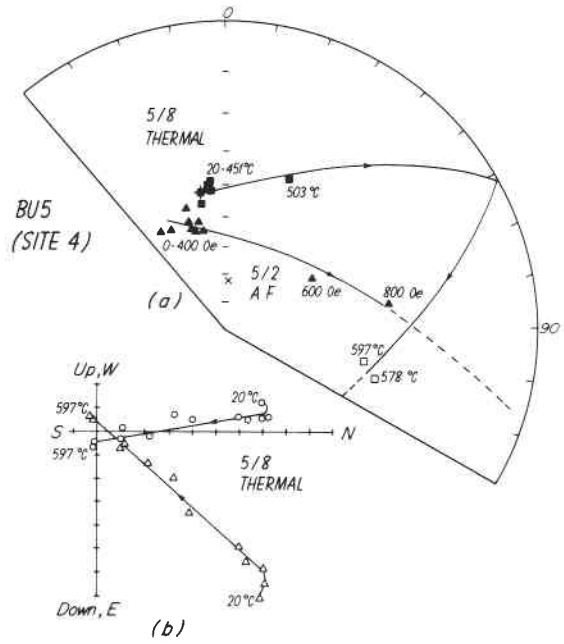


FIG. 3. Stereoplot and vector diagram of demagnetization behaviour of sample BU5, whose dominant NRM is 1N. The 1N vector is 15−25° shallower than the PEF, which is represented by X.

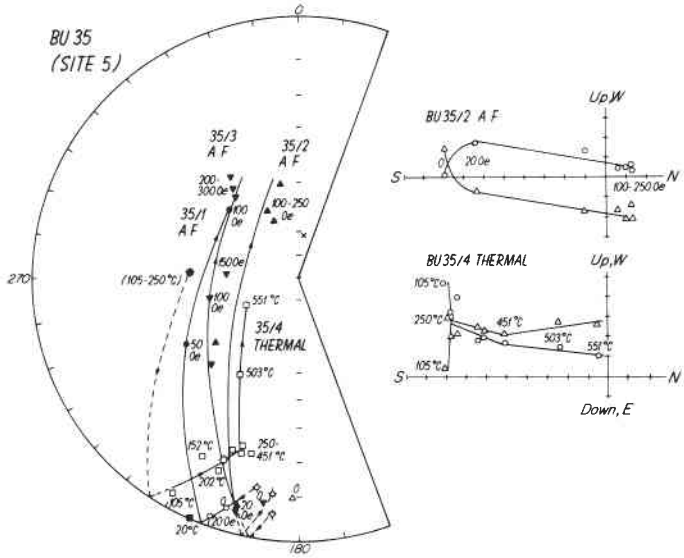


FIG. 4. Complete (AF) and partial (thermal) 1R → 1N reversals of the NRM vector of sample BU35 in the course of demagnetization. A minor PEF overprint is evident in the AF and thermal vector plots from 0 to 20 Oe and 105 to 250°C, respectively, but the main component revealed is 1R.

be isolated.

Almost complete 1R → 1N reversals are revealed by AF and thermal demagnetization of sample BU35 (Fig. 4). The 1R vector has very low H_C's (AF vector diagram) but its T_B's span the range 250−≥551°C. The carrier is likely coarse MD magnetite. There is also a small, recent, viscous overprint revealed between 20 and 250°C. The 1N vector is not isolated by thermal cleaning and no conclusions can be drawn about its T_B range. It is certainly a minor component compared with 1R in this sample.

The NRM of sample BU57 (Fig. 5) is a superposition of either 1R + 1N or of 1R + recent viscous remanent magnetization (VRM) directed near the PEF. Thermal cleaning suggests a PEF overprint, because the blocking temperatures do not exceed 250°C (Wilson and Smith 1968; Dunlop 1969). However, on AF treatment, although the bulk of the northerly, downward remanence is erased between 20 and 200 Oe (2 and 20 mT), the coercivity spectrum extends to 600 Oe (60 mT) (Fig. 5*b*). Such H_C's are unusually high for a VRM, unless carried by hematite. It is possible that the NRM contains, as well as 1R, an unresolved sum of 1N and a PEF overprint.

The 1R remanence of BU57 has T_B's of 250−659°C. It is carried by both magnetite and hematite, but principally the former.

In summary, the characteristic NRM of the Burchell Lake granite is a type 1 vector of either normal or reverse polarity. The predominant remanence is 1N in 21 of the samples and 1R in the other 15. In samples that carry both polarities of NRM, the temperature at which one is erased and the other isolated is variable and does not correspond to the magnetite Curie point. These observations are consistent with the NRM being a TRM acquired in initial cooling, although they do not prove that this is so.

Mean directions from stable end points, subtracted vectors, and remagnetization circles

A three-tier statistical analysis of type 1 directions from 36 samples appears in Table 1. The 17 samples not included in the analysis were either weakly and unstably magnetized or lightning struck (see paper III).

The mean type 1 direction is $D = 2.3°$ $I = 48.9°$ ($k = 23.9$, $\alpha_{95} = 10.0°$, $N = 10$ sites). The number of stable samples per site varies considerably, but giving unit weight to samples results in a very similar mean: $D = 3.6°$ $I = 49.8°$. These directions are similar to but slightly shallower than the 1N + 1R mean, $D = 7.5°$ $I = 55.4°$, for the Shelley Lake granite (paper II). It is unmistakably distinct from both PEF and axial dipole field directions at the 95% confidence level (Fig. 6).

An estimate of the mean type 1 direction can also be obtained by projecting incomplete great-circle vector swings (remagnetization circles, Halls 1976) to their common point of intersection near 1R or 1N. Some of these trajectories are shown in Fig. 7*a*, and the normals to the remagnetization planes represented by the great-circle traces are plotted in Fig. 7*b*. The normal to the plane defined by the best fitting great circle through all 25 remagnetization normals is the intersection point of the original trajectories.

The mean intersection point is $D = 176.3°$ $I = -42.1°$, with precision k' (equivalent to the Fisher k for dispersed unit vectors) of 17.9 and confidence oval semi-axes $\beta_{95} = 10.2°$, $\gamma_{95} = 11.8°$. At the 95% confidence level, this estimate of the mean 1R direction is indistinguishable from the mean of stable end points (cf. Fig. 7*b*).

The comparative circularity of the confidence oval reflects the widely dispersed azimuths of remagnetization circles, which in turn indicate that 1N and 1R directions are very nearly 180° reversed to each other. The N mean is about 10° steeper than the R mean (Table 1), perhaps because of some unresolved PEF contamination of the former, but the two means are nevertheless antipodal within the limits of their 95% confidence circles.

Type 2 NRM results

Twelve of the Burchell Lake samples possess a type 2 NRM component, with shallow inclination and either east to north-

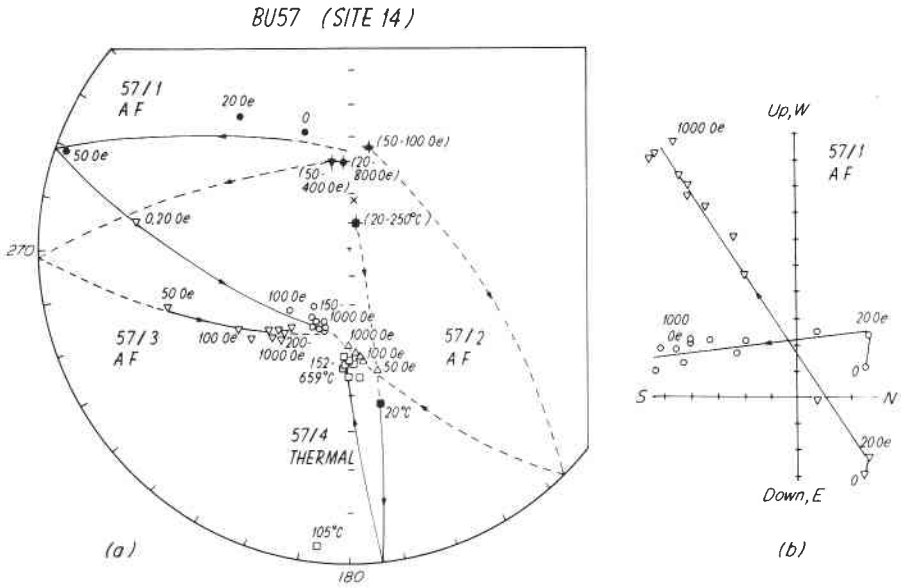


FIG. 5. Underlying high-H_C, high-T_B 1R remanence revealed by AF and thermal cleaning of sample BU57. The low-T_B component has a direction and T_B spectrum appropriate to a viscous PEF overprint, but the AF vector diagram indicates some H_C's up to 600 Oe. A 1N remanence and a PEF overprint may be superimposed in the starting remanence.

TABLE 1. Burchell Lake granite, type 1 site-mean directions after AF or thermal cleaning

Site	N	D (°)	I (°)	k	α_{95} (°)
1	7	15	47.5	12.9	17.4
4	5	9	50.5	29.3	14.3
5	4	4	28	28.8	17.4
7	2	344	65	602	10.1
9	2	343	56	29.2	47.9
10	1	15	21.5		
11	3	346	47	21.3	27.3
12	4	1	65	35.4	15.6
13	3	9	49	32.8	21.8
14	5	348	51.5	19.5	17.7
Mean, unit weight to samples	36	3.6	49.8	16.3	6.0
Mean, normal samples	21	1.3	55.6	28.7	6.3
Mean, reverse samples	15	185.7	−44.4	12.1	9.8
Mean, unit weight to sites	10	2.3	48.9	23.9	10.0
Paleopole		83.2E	71.1N		
		($\delta p = 10.3°$, $\delta m = 14.2°$)			

NOTES: N is the number of samples or sites averaged; D, I are declination and inclination of magnetization; k is the Fisher precision parameter; and α_{95} is the radius of the cone of 95% confidence about the mean direction.

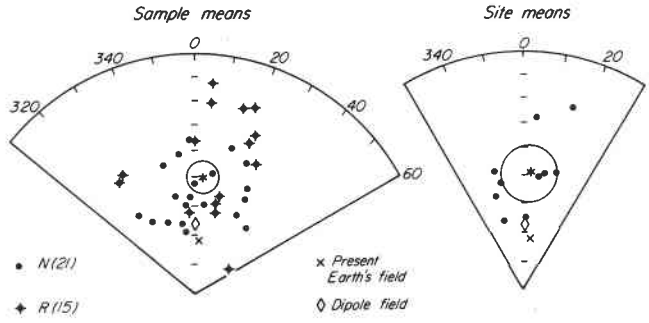


FIG. 6. Equal-area projections of AF and thermally cleaned sample-mean and site-mean type 1 magnetizations for the Burchell Lake granite. The grand means are indicated by stars surrounded by circles of 95% confidence.

east declination ("normal" polarity, 2N) or west to southwest declination ("reverse" polarity, 2R). As with the type 1 NRM, the polarities are arbitrary, because the apparent polar wander path (APWP) for North America in the Precambrian has gaps over several fairly long time intervals. Type 2 remanences are rarely isolated as stable end-point vectors by AF or thermal cleaning. Typically they are revealed by non-great-circle vector trajectories on stereonets or by linear segments in orthogonal vector projections.

Sample BU56 (Fig. 8) is a complicated but particularly interesting example. Between 20 and 202°C, a PEF overprint is progressively erased and the resultant NRM vector migrates westward. From 202 to 551°C, the resultant moves towards 1R with the removal of a vector near $D = 255°$ $I = 10°$. This latter vector is a type 2 remanence. Similarly directed NRM's were reported for the Wabigoon gabbro and the Shelley Lake granite (papers I and II).

The method of Hoffman and Day (1978) as modified by Halls (1979) was also used to estimate the type 2 direction in BU56. The great circle through high-T_B resultant vectors and the great circle through low-T_B *difference* vectors (subtracted vectors between adjacent temperature steps, represented in Fig. 8 by squares with diagonal crosses) were projected to their point of intersection. This estimate of the type 2 direction is consistent with the direction obtained from the vector plot (Fig. 8 *a* and *b*) but is less accurate because the difference vectors are very scattered compared with resultant vectors.

An independent estimate of the 2R direction was obtained from the AF data between 50 and 200 Oe (5 and 20 mT) (Fig. 8 *a* and *c*). In this sample, 2R is the softest remanence. From 200 to 800 Oe (20 to 80 mT) a 1N component is eliminated (Fig. 8 *a* and *c*) and above 800 Oe (80 mT) the 1R vector is isolated as a stable end point.

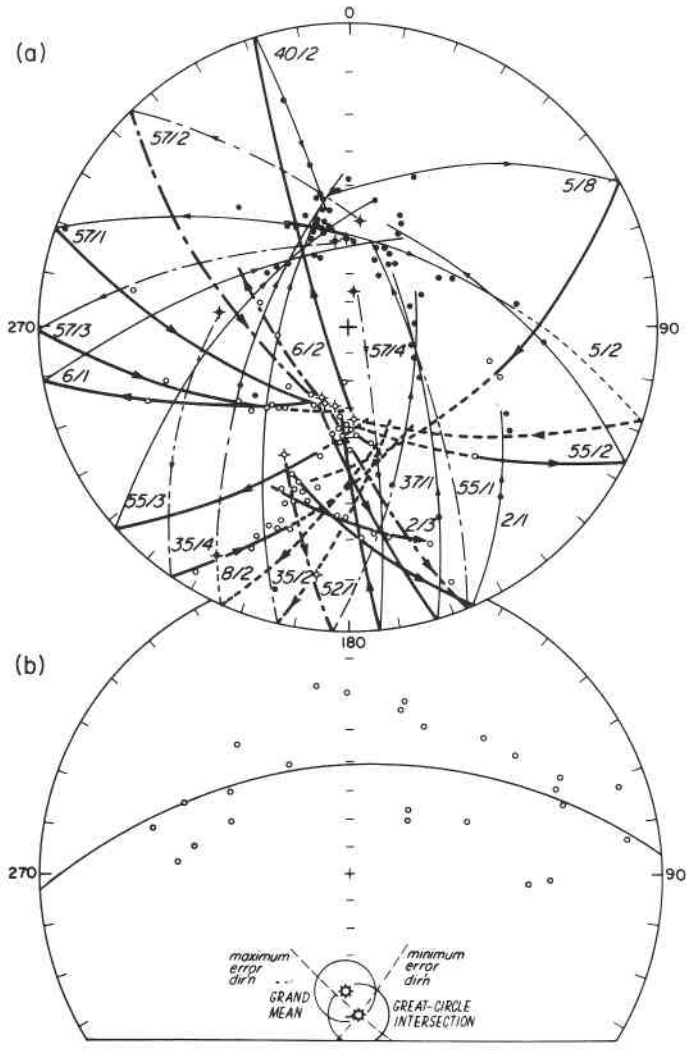


FIG. 7. (*a*) Stereographic projection of typical demagnetization trajectories of NRM vectors in the course of AF or thermal cleaning for the Burchell Lake granite. The trajectories, if projected, tend to intersect near 1R or 1N. (*b*) Stereoplot of the normals to remagnetization planes represented by the great-circle trajectories in (*a*), the best fitting great circle through the normals, and the normal to the plane represented by this great circle, which is the best estimate of the intersection point of the original trajectories (see Halls 1976). Within the 95% confidence limits shown, the intersection point agrees with the 1R grand-mean direction.

The results of a two-tier statistical analysis of type 2 subtracted vectors appears in Table 3. The mean for $N = 12$ samples is $D = 80.9°$ $I = -11.1°$, but the precision is low ($k = 7.7$). There is little doubt that type 2 remanence is present in some Burchell Lake samples, but the subtracted vectors are so noisy that the mean is not defined at an acceptable level of confidence.

Paleopoles and apparent polar wander

The paleomagnetic pole BL1 corresponding to the mean type 1 direction (Table 1) falls at 83.2°E, 71.1°N, near the late Archean APWP for Laurentia (Fig. 9). Its position is intermediate between pole CS of the 2570 ± 170 Ma Chibougamau sills (Ueno and Irving 1976) and pole SL1 for the 2580 ± 20 Ma Shelley Lake granite type 1 NRM, which is argued in II to be a primary TRM.

TABLE 2. Burchell Lake granite, type 1 directions from remagnetization circles intersecting near 1R

	N	*D* (°)	*I* (°)	*k'*	β_{95} (°)	γ_{95} (°)
1R, unit weight to specimen normals	25	176.3	−42.1	17.9	10.2	11.8
Paleopole		97.5E	65.5N			
		($\delta p < 9.4°$, $\delta m < 14.5°$)				

NOTES: The parameter k' corresponds to the Fisher k of Table 1 and the confidence oval semi-axes β_{95} and γ_{95} are analogous to α_{95}.

TABLE 3. Burchell Lake granite, type 2 mean direction after AF or thermal clearing

	N	*D* (°)	*I* (°)	*k*	α_{95} (°)
Mean, unit weight to samples	12	80.9	−11.1	7.7	16.7
Paleopole		170.1W	1.7S		
		($\delta p = 8.4°$, $\delta m = 16.7°$)			

NOTE: Symbols as in Table 1.

The Burchell Lake body is not dated radiometrically but, as outlined earlier, biotite K/Ar ages for other intrusions in the immediate area are between 2550 and 2610 Ma (Giblin 1964, p. 21). The age implied by the paleomagnetic results is therefore not surprising, although BL1 certainly cannot be viewed as a tie-point pole for the APWP.

The Burchell Lake type 2 remanence is too scattered to define a reliable mean and the corresponding paleopole (Table 3) is untrustworthy.

Discussion

Since the 1N and 1R remanences occur with about equal frequency, are almost exactly 180° reversed to each other, and have similar H_C and T_B spectra, they were presumably produced by the same mechanism but at times sufficiently different to encompass a geomagnetic field reversal. Whether the mechanism was TRM, thermochemical remanent magnetization (TCRM), or chemical remanent magnetization (CRM) is uncertain. A contact test was attempted but it failed because the reheated country rocks (gabbro, rhyolite, amphibolite schist) gave generally unstable and inconsistent paleomagnetic results. The other possible test for a primary TRM, a correlation between NRM polarity and radial movement of a cooling front through the intrusion (cf. paper I), was inconclusive because all samples came from marginal sites.

In multivectorial NRM's, thermal overprints have T_B spectra with little or no mutual overlap and hence yield ideal linear segments on vector diagrams (e.g., Dunlop 1979), whereas chemical or thermochemical overprints can have considerable overlap in blocking temperature with the remanences they overprint. Indeed, if the CRM is carried by hematite, its T_B's will be higher than those of the older magnetite NRM. Burchell Lake samples have apparently nonoverlapping 1N and 1R T_B spectra (Figs. 3, 4), and the temperature separating the two spectra is usually different from the magnetite Curie temperatures indicated by thermomagnetic curves as in Fig. 1.

Unfortunately, in the case of antiparallel vectors, it is not easy to be sure that the two vectors occupy mutually exclusive T_B ranges. Sample BU2 (Fig. 2), in which the resultant vector swung away from 1R below 503°C but returned towards 1R at higher temperatures, may be an illustration of complicated

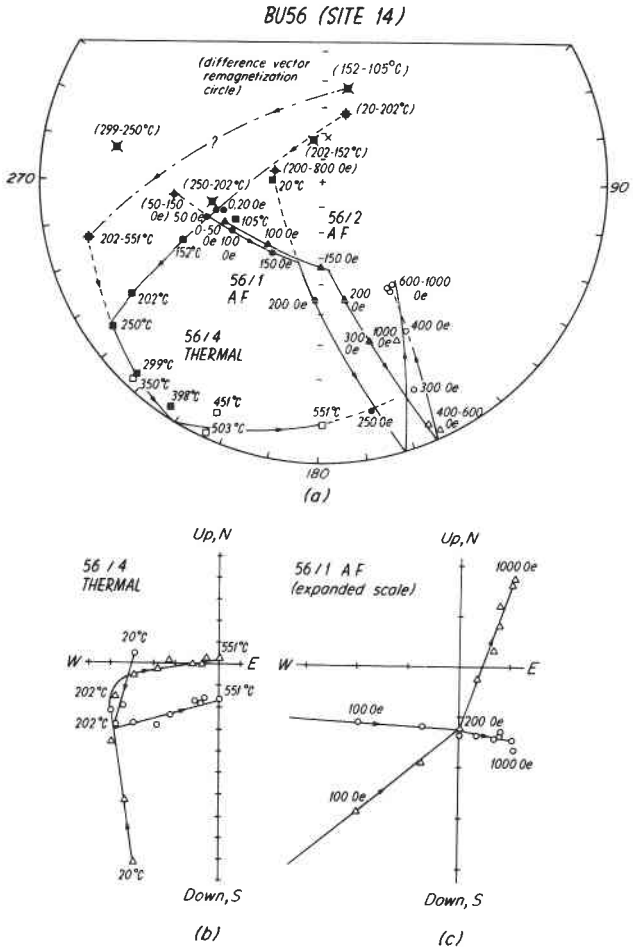


FIG. 8. PEF + 2R + 1R magnetizations in sample BU56. Points with diagonal crosses represent difference vectors between adjacent thermal demagnetization steps (299−250, 250−202, 202−152, 152−105°C). The remagnetization circle through these difference vectors (dash−dot line) intersects the remagnetization circle through high-T_B resultant vectors (299, 350, . . . , 551°C) near the subtracted vector removed between 202 and 551°C, which is the best estimate of the 2R remanence. An independent estimate of 2R is given by the 50−150 Oe subtracted vector for 56/1 (*a*, *c*).

overlap of the 1R and 1N T_B spectra. If so, both NRM's are probably CRM's or TCRM's rather than TRM's, and no simple pattern of polarities within and between sites would be expected.

Based on rock magnetic properties, the NRM's are more likely to be CRM's than TRM's because of their very low intensities compared with saturation IRM.

Whether the type 1 NRM is a TRM or a CRM/TCRM, it was probably acquired within at most 50 Ma of initial cooling of the Burchell Lake stock, because pole position BL1 is in good agreement with ≈2600 Ma paleopoles from track 6 of the Laurentian APWP (Irving 1979). Since BL1 is not independently dated, however, it cannot be used as a primary datum anchoring the APWP.

Conclusions

The characteristic NRM of the Burchell Lake granite defines a paleopole at 83.2°E, 71.1°N (δp = 10.3°, δm = 14.2°), around 2600 Ma on the Archean APWP for North America. The Burchell Lake pluton is likely 2550−2600 Ma in age, and

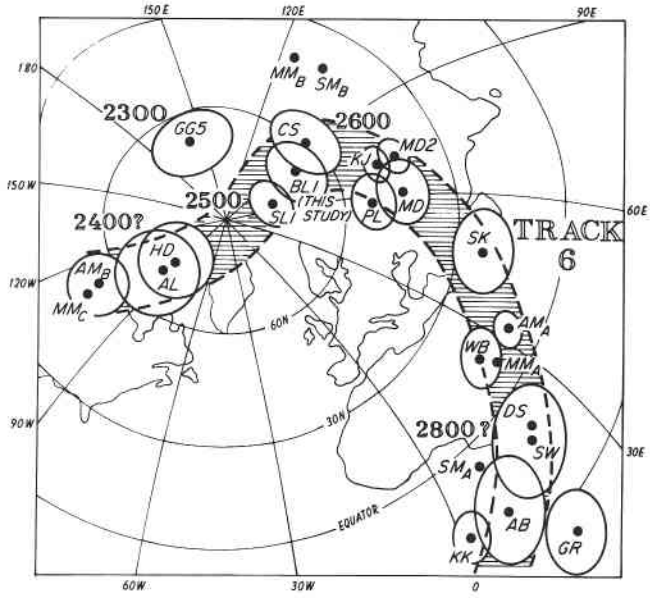


FIG. 9. Tentative late Archean APWP for Laurentia. The various primary and secondary paleopoles are identified in Irving (1979, Fig. 11) and Dunlop (1984, Fig. 12). The position of pole BL1 for the type 1 NRM of the Burchell Lake granite is consistent with magnetization soon after intrusion at 2550−2600 Ma.

so the characteristic NRM was probably acquired soon after formation of the body. However, the age of magnetization has not been independently established.

Type 2 NRM, probably of secondary origin, is present in some Burchell Lake samples, but is too scattered in direction to yield a reliable pole position.

Acknowledgments

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BAILEY, M. E., and DUNLOP, D. J. 1983. Alternating field characteristics of pseudo-single-domain (2−14 μm) and multidomain magnetite. Earth and Planetary Science Letters, **63**, pp. 335−352.

BAILEY, R. C., and HALLS, H. C. 1984. Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data. Journal of Geophysics. (In press.)

DUNLOP, D. J. 1969. Comments on a paper by R. L. Wilson and P. J. Smith: The nature of secondary magnetizations in some igneous and baked rocks. Journal of Geomagnetism and Geoelectricity, **21**, pp. 797−799.

——— 1979. On the use of Zijderveld vector diagrams in multicomponent paleomagnetic studies. Physics of the Earth and Planetary Interiors, **20**, pp. 12−24.

——— 1983. Paleomagnetism of Archean rocks from northwestern Ontario: Wabigoon gabbro, Wabigoon Subprovince. Canadian Journal of Earth Sciences, **20**, pp. 1805−1817.

——— 1984. Paleomagnetism of Archean rocks from northwestern

Ontario: II. Shelley Lake granite, Quetico Subprovince. Canadian Journal of Earth Sciences, **21**, pp. 869–878.

DUNLOP, D. J., SCHUTTS, L. D., and HALE, C. J. 1984. Paleomagnetism of Archean rocks from northwestern Ontario: III. Rock magnetism of the Shelley Lake granite. Canadian Journal of Earth Sciences, **21**, pp. 879–886.

GIBLIN, P. E. 1964. Geology of the Burchell Lake area, District of Thunder Bay. Ontario Department of Mines, Geological Report No. 19, 39 p. and Geological Map No. 2036, scale 1 in. = $\frac{1}{2}$ mi.

HALLS, H. C. 1976. A least-squares method to find a remanence direction from converging remagnetization circles. Geophysical Journal of the Royal Astronomical Society, **45**, pp. 297–304.

——— 1979. Separation of multicomponent NRM: combined use of difference and resultant magnetization vectors. Earth and Planetary Science Letters, **43**, pp. 303–308.

HOFFMAN, K. A., and DAY, R. 1978. Separation of multicomponent NRM: a general method. Earth and Planetary Science Letters, **40**, pp. 433–438.

IRVING, E. 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terrains. Canadian Journal of Earth Sciences, **16**, pp. 669–694.

LOWRIE, W., and FULLER, M. 1971. On the alternating field demagnetization characteristics of multidomain thermoremanent magnetization in magnetite. Journal of Geophysical Research, **76**, pp. 6339–6349.

UENO, H., and IRVING, E. 1976. Paleomagnetism of the Chibougamau Greenstone Belt, Quebec and the effects of Grenvillian post-orogenic uplift. Precambrian Research, **3**, pp. 305–315.

WILSON, R. L., and SMITH, P. J. 1968. The nature of secondary magnetizations in some igneous and baked rocks. Journal of Geomagnetism and Geoelectricity, **20**, pp. 367–380.